

1.2 Lesson: From a Single Neuron to a Neural Network



From a Single Neuron to a Neural Network

A single neuron performs a simple transformation, but deep learning derives its power from **layered networks of neurons**. A **layer** consists of a row of neurons, each having its own weight vector but sharing the input vector:

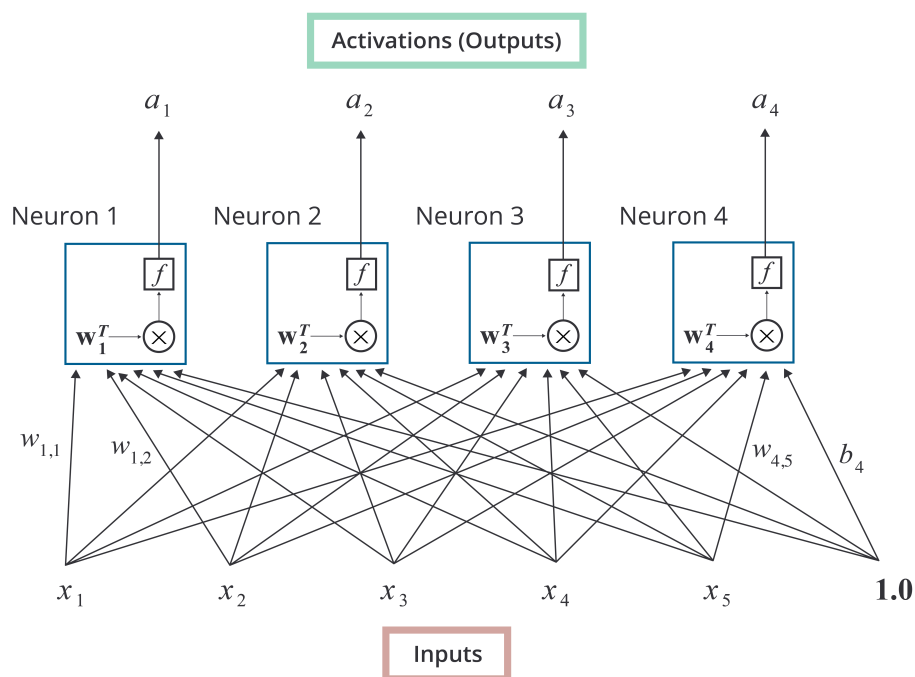


Figure 1: Single Network Layer

In linear algebra terms, we are simply doing a matrix multiplication and then applying the activation function f :

$$W \mathbf{x} = \begin{bmatrix} w_{1,1} & w_{1,2} & w_{1,3} & w_{1,4} & w_{1,5} & b_1 \\ w_{2,1} & w_{2,2} & w_{2,3} & w_{2,4} & w_{2,5} & b_2 \\ w_{3,1} & w_{3,2} & w_{3,3} & w_{3,4} & w_{3,5} & b_3 \\ w_{4,1} & w_{4,2} & w_{4,3} & w_{4,4} & w_{4,5} & b_4 \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_2 \\ x_4 \\ x_5 \\ 1.0 \end{bmatrix} = \begin{bmatrix} \mathbf{w}_1^T \mathbf{x} \\ \mathbf{w}_2^T \mathbf{x} \\ \vdots \\ \mathbf{w}_n^T \mathbf{x} \end{bmatrix}$$

$$f \begin{pmatrix} \mathbf{w}_1^T \mathbf{x} \\ \mathbf{w}_2^T \mathbf{x} \\ \mathbf{w}_3^T \mathbf{x} \\ \mathbf{w}_4^T \mathbf{x} \end{pmatrix} = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \\ \mathbf{a}_4 \end{bmatrix}$$

$$\mathbf{a} = f(W\mathbf{x})$$

Figure 2: Linear Algebra Formula for Network Layer

In a **fully-connected feed-forward network (FFNN)**, each layer is connected to each neuron or input in the previous layer.

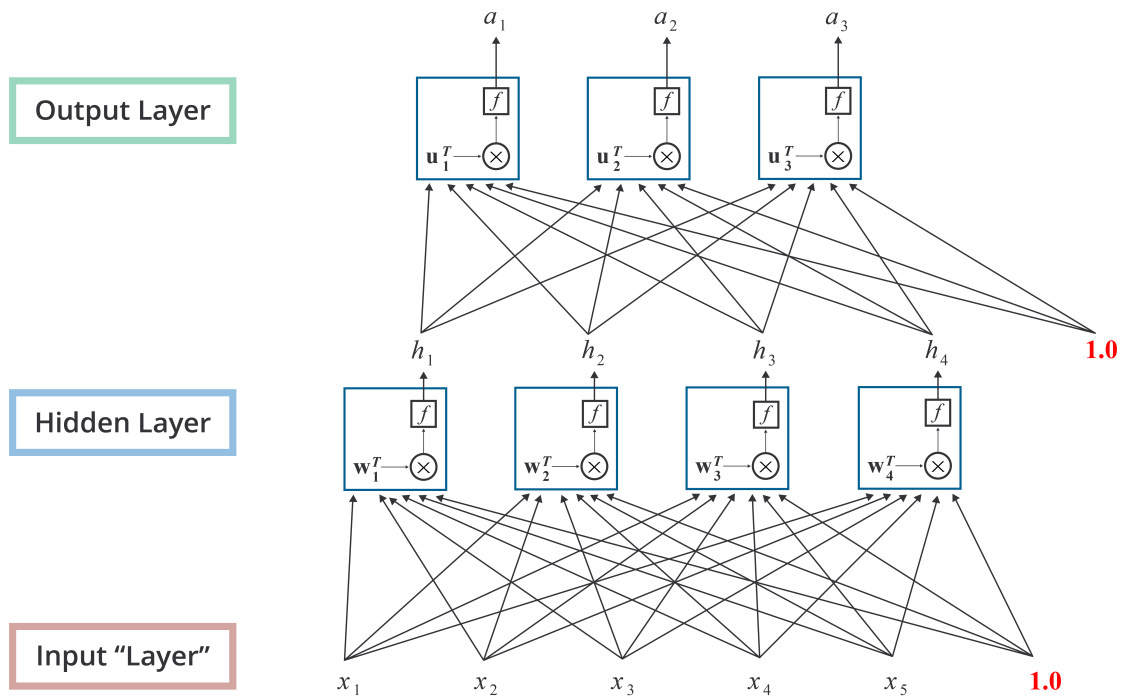


Figure 3: Network with One Hidden Layer

A **deep learning model** is defined by its **parameters** and **hyperparameters**:

- **Parameters** are the weights for all the neurons in the network, adjusted during training for some task.
- **Hyperparameters** are choices about:
 - How many layers
 - How many neurons in each layer
 - Which activation functions in each layer
 - Training algorithms (which we cover in Week 2)

Your choice of hyperparameters defines the structure of your network, but parameters contain the actual information content of the model.